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Application of the Progressively Lowered Stress Threshold Model with Community-Based Caregivers

A Randomized Controlled Trial



ABSTRACT

The Progressively Lowered Stress Threshold (PLST) is a conceptual model for reducing behavioral symptoms in individuals with dementia. The aim of the current study was to evaluate the effectiveness of the PLST-based intervention on burden, symptoms of depression, and quality of life (QOL) of caregivers, as well as neuropsychiatric symptoms and QOL of individuals with dementia. A randomized controlled trial was used. Sixty-five participants completed the study. Although variables examined in caregivers in the intervention group showed statistically significant improvement throughout follow up, there was no statistically significant difference between the intervention and comparison groups. In addition, no statistically significant difference was noted between groups for patient variables. Care based on the PLST model was clinically effective in decreasing caregiver burden and depression of family members, and in increasing their QOL, but no more effective than routine care. [*Journal of Gerontological Nursing*, 42(7), 44-54.]

The incidence and prevalence of dementia are on the rise due to demographic trends leading to rapidly aging populations across the globe. Worldwide, 47.5 million individuals have dementia and there are 7.7 million new cases every year (World Health Organization [WHO], 2015). The total number of individuals with dementia is projected to reach 75.6 million in 2030 and almost triple by 2050 to 135.5 million (WHO, 2015). The number of individuals with dementia is also increasing in Turkey. Although there are no current prevalence studies in Turkey, according to a study conducted in 2009, dementia prevalence was 8.4%, which increases parallel to age; the rate of incidence was 2.2% for individuals ages 55 to 59 and 30.4% for those 75 and older (Arslantas et al., 2009).

Individuals with dementia present many difficult care problems because of the progressive deterioration of the cerebral cortex, which results in memory loss, functional decline, loss of the ability to participate socially, and a bewildering array of behaviors. Care needs of individuals with dementia typically increase over time and most even-

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tually become fully reliant on their caregivers (Jennings et al., 2015; Zhu et al., 2015).

Caregiving can become an all-consuming job. For many families, providing care is fraught with difficulties and caregivers may spend a significant amount of time trying to provide care recipients with a sense of connectedness while anticipating their next move. Families often refuse long-term care options until they are desperate and care needs are overwhelming. Similarly, in Turkey, individuals and society in general find it culturally difficult to accept institutional care because Turkish culture and society traditionally promote strong family bonds and familial care for those in need. Because structured home care services are not sufficient in Turkey, families try to continue home care without support.

Behavioral changes and caregiver stress or illness are the primary reasons for institutionalizing individuals with Alzheimer's disease. Some studies have concluded that behavioral changes are most problematic for caregivers (Cerejeira, Lagarto, & Mukaetova-Ladinska, 2012; Cheung et al., 2015). More than 90% of all individuals with dementia will experience behavioral and psychological symptoms, including anxiety, apathy, agitation, and depression (Brodaty & Arasaratnam, 2012; Cerejeira et al., 2012; Laks, Goren, Dueñas, Novick, & Wroblewski, 2015). When caregivers lack the skills to manage these problems effectively, the disturbing behaviors often escalate, leading to increased confusion, agitation, and a downward spiral into excessive disability. Management of these symptoms can be challenging. Negative outcomes, such as an increased burden and depression associated with caregiving, are well documented (Akpınar, Küçükgülü, & Yener, 2011; Crellin, Orrell, McDermott, & Charlesworth, 2014; Laks et al., 2015).

Behavioral and psychological symptoms of dementia are often managed with pharmacological interventions, such as antipsychotic medications. These medications may offer some reduction in symptoms (Sink, Holden, & Yaffe, 2005), but pharmacological therapies have a number of potential adverse side effects, such as increased falls and mortality (Ballard & Howard, 2006), and a decreased quality of life (QOL) for individuals with dementia (Woods, Craven, & Whitney, 2005). For these reasons, first-line treatment involves observation of symptoms of dementia (Laks et al., 2015) and focuses on the use of nonpharmacological interventions, such as music, massage, therapeutic touch, pet therapy, and aromatherapy (Brodaty & Arasaratnam, 2012; Capezuti, Zwicker, Mezey, & Fulmer, 2008; Pinquart & Sörensen, 2006).

Using these nonpharmacological interventions, behaviors can often be managed while eliminating the need for physical or chemical restraints (Brodaty & Arasaratnam, 2012; Cohen-Mansfield, Libin, & Marx, 2007). Psychoeducational interventions in particular are thought to have a beneficial and reassuring effect on caregivers (Chien & Lee, 2011; Ducharme, Lachance, Lévesque, Zarit, & Kergoat, 2015).

Studies have provided preliminary support for the use of this approach in decreasing caregiver burden and symptoms of depression, and in increasing the QOL of caregivers. However, most studies have only reported short-term results among caregivers (Sörensen, Pinquart, Habil, & Duberstein, 2002) and there is a clear need for further investigation. In addition, little research has evaluated nursing interventions that adhere to models, are theoretically driven, and may be clinically useful in dementia. The use of conceptual frameworks allows for the organization of complex material into

simpler, more manageable components for planning and evaluating care. Mitty and Flores (2007) reported that the Progressively Lowered Stress Threshold (PLST) model is an evidence-based model that can be used in dementia.

The aim of the current study was to evaluate the effectiveness of a community-based training program designed to teach non-professional caregivers to manage the behavioral problems of individuals with Alzheimer's disease using the PLST-based intervention, and to compare this training program with a routine care program.

In accordance with this aim, the following hypotheses were tested:

1. The PLST-based intervention will be more effective than routine care in reducing burden and depression, and in increasing QOL of caregivers of individuals with dementia.
2. The PLST-based intervention will be more effective than routine care in reducing distressing behavioral symptoms and in increasing QOL for individuals with dementia.

PROGRESSIVELY LOWERED STRESS THRESHOLD MODEL

The PLST model (Hall & Buckwalter, 1987) provides the foundation for a psychoeducational intervention that assists formal and informal caregivers in understanding behaviors and planning care for individuals with dementia. It has been tested widely since 1988. The model can be easily understood by professional and non-professional caregivers and allows for flexible modification of the care plan based on individuals' changing needs.

The PLST model is a conceptual framework that organizes symptoms, suggests causes of challenging behaviors, and teaches caregivers methods of modifying the environment to eliminate or minimize behavioral symptoms. The reasons for, or triggers of, symptoms

can usually be modified in any setting, resulting in a decrease in behaviors distressing to caregivers and individuals with dementia. The PLST model should empower caregivers by helping them prevent and manage the most difficult behaviors in Alzheimer's disease. The model is based on patients' reduced tolerance to stress, reflected in a higher proportion of altered behavior states, which occurs during the course of the disease. As the stress threshold declines, there is a progressive reduction in normative behaviors and an increase in anxious and dysfunctional behaviors. An important assumption of the PLST model is that all stress-related behavior has meaning and an underlying cause (Hall &

METHOD

Design and Sample

A randomized controlled trial with a repeated-measures design was used. The study is reported according to CONSORT 2010 statement guidelines (Schulz, Altman, & Moher, 2010). Baseline and posttest assessment screenings were conducted and interventions were delivered during home visits. The researchers were knowledgeable about application of PLST, and treatment fidelity was evaluated based on study design, training of interventionists, and delivery and receipt of the intervention (Bellg et al., 2004). There was strong support for all areas. However, there were some concerns about receipt of the intervention. Using the model in the com-

(0.99) and effect size (0.5), with an alpha value of 0.05, was evaluated using NCSS 2004 and PASS 2002. Generally, a power of 0.8 is acceptable for such studies (Akgül, 2005). Statistical power was calculated before data collection to decide the sample needed for the current study; based on average scores from Bateman and Fonagy (1999), 20 participants were needed for each group. To decide the final sample, possible losses were considered, so the final sample included 35 participants in each group.

Participants were recruited between October 2012 and October 2013 in a large city in western Turkey. The written aims of the study, instructions, and time required for the measures to be completed were communicated to participants. Participants were informed of the purpose of the study, and confidentiality of their personal identity and data was guaranteed. Participants were also assured of their right to terminate participation at any time. Patient lists were obtained from dementia outpatient clinics at one university hospital and the office of the Alzheimer's Association in Izmir. Potential participants and their family members were then identified.

After obtaining caregivers' consent, individuals with dementia were screened and considered eligible if they met the following criteria: diagnosed with Alzheimer's-type dementia, demonstrated signs of early and mild dementia, and had volunteered to participate in the study. Inclusion criteria for caregivers were: being a close relative of the patient, being the primary caregiver, having volunteered to participate in the study, and providing ≥ 4 hours of supervision or care weekly. Older caregivers often consult with adult children and other family members about whether to adhere to advice given by professionals. However, for purposes of the current study, data

The Progressively Lowered Stress Threshold model should empower caregivers by helping them prevent and manage the most difficult behaviors in Alzheimer's disease.

Buckwalter, 1987). The model identifies the following six factors as contributing to stress in individuals with dementia: (a) physical stressors (e.g., infection, pain); (b) misleading or inappropriate stimuli; (c) change of environment, caregiver, or routine; (d) internal or external demands that exceed functional capacity; (e) fatigue; and (f) affective responses to perceptions of loss (Hall & Buckwalter, 1987; Hall et al., 1995).

Studies on PLST-based interventions to reduce behavioral problems with dementia have been conducted mainly in nursing homes, but the current study was conducted in the community. Using the model in the community is more challenging because the level of environmental manipulation that an institution can achieve might not be possible in a home.

community was more challenging because of the cultural expectations of privacy in the domestic life in Turkey. Researchers were unable to perform observations as to whether the care plan was effectively applied by caregivers of the patient in the home. Thus, data relied on caregiver feedback.

The sample comprised individuals with dementia and their caregivers. Although a sample of 70 participants was planned, 65 participants completed the study. In the experimental group, two participants were lost to follow up (one patient was diagnosed with stomach cancer and his caregiver decided to discontinue participation, and the other patient moved out of town). In the comparison group, three participants were lost to follow up (two patients moved out of town and one was unwilling to continue). The power of the study

TABLE 1
PROTOCOLS OF THE EXPERIMENTAL AND COMPARISON GROUPS

Session	Experimental Group (PLST Intervention)	Comparison Group (Routine Care)
1	Meet the patient and caregiver	Meet the patient and caregiver
	Review aims and protocols of the study	Review aims and protocols of the study
	Obtain informed consent	Obtain informed consent
	Data collection	Data collection
	Collect information about symptoms and coping strategies	
	Collect detailed information about the financial situation, domestic conditions, and familial support available	
	Briefly describe PLST theory, concept, and plan of care	
	Briefly share information about stimuli that may cause stress for the patient	
	Supply researchers' telephone numbers	
	Set the date of the next interview	
2	Review principles/concepts/plan of care of PLST model and answer questions	Supply general information describing care of individuals with dementia
	Share information about social services provided for the care of individuals with Alzheimer's disease	Set the date of the next interview
	Make phone calls every 2 weeks and answer caregivers' questions about the application of the care plan being prepared according to PLST	
	Set the date of the next interview	
3	Data collection	Data collection
	Develop a care plan aimed at solving any new problems	
4	Data collection	Data collection

Note. PLST = Progressively Lowered Stress Threshold.

were collected only from the identified primary caregivers.

Approval to conduct the study was obtained from the Ethical Committee of the Institute of Health Sciences, Dokuz Eylul University, Turkey.

Randomization and Blinding

The sample initially comprised 70 individuals with dementia and their family caregivers (who took care of them at home). Thirty-five (50%) participants were randomly assigned to the PLST-based intervention (experimental group) and 35 (50%) were assigned to the routine care program (comparison group) (Akgül, 2005). There were no statistically significant differences by group between individuals with dementia and their caregivers in terms of sociodemographic

characteristics and dependent variables at baseline.

Intervention

The experimental group ($n = 35$) received the PLST-based intervention and the comparison group ($n = 35$) received a routine care program. Interventions were conducted at patients' homes in their usual environment. Individuals with dementia and their caregivers in the experimental group were seen four times in a 6-month period. Each session lasted at least 1 hour. The intervention phase lasted 3 months. Follow-up measures were performed 3 months later after the third home visit. Protocols for the experimental and comparison groups are shown in Table 1. All

home visits were conducted at the same time interval for both groups.

The first home visit focused on understanding the behaviors of the individuals with dementia, how caregivers coped with these behaviors, and difficulties experienced by caregivers. After 2 weeks, during the second home visit for caregivers in the experimental group, the PLST-based intervention was provided and demonstrated. At the third home visit 2 months later, the care plan was reviewed, more techniques were taught, and written materials summarizing the care plan were provided. After the third home visit, follow-up telephone calls were made every 2 weeks. Experimental group members also received a caregiver booklet prepared

by the authors that contained a variety of educational and support materials, such as general information about the disease, common problems (e.g., sleep disturbances), how to administer medications, bathing, toileting, physical activities, nutrition, and communication, as well as recommended reading for family members caring for individuals with dementia.

Comparison group members received routine care. In the routine care program, following diagnosis, caregivers were informed about the disease by physicians and given leaflets. When caregivers came to the outpatient visit with the patients, they had an opportunity to ask physicians questions. Frequency of outpatient visits varied between every 6 months and once per year. Participants in the comparison group were seen four times in a 6-month period at their home. Each session lasted approximately 1 hour. The first home visit focused on a review of the study aims and protocols, obtaining informed consent, and data collection. After 2 weeks, during the second home visit for caregivers in the comparison group, general information about care of individuals with dementia was presented and updated leaflets were provided. At the third home visit 2 months later, data were collected. At the last home visit 3 months later, data were again collected. After the study was completed, participants in the comparison group were provided with the experimental group protocols.

Measures

Caregivers were assessed on three outcome measures at baseline (within 2 weeks prior to commencement of the intervention) and at the 3- and 6-month visits: (a) caregiver burden; (b) depression; and (c) QOL for individuals with dementia-related behavioral problems. Family caregivers were asked

to complete the Turkish versions of scales for pre- and posttesting to detect effects of the PLST-based intervention. Individuals with dementia were asked to complete only the QOL-Alzheimer's disease (QOL-AD) scales. Demographic data were collected at baseline. The instruments required approximately 35 minutes for completion.

Caregiver Burden Inventory. The Caregiver Burden Inventory (CBI) is a 24-item scale, where higher scores indicate greater caregiver burden (Novak & Guest, 1989). Validity and reliability of the CBI for the Turkish population was studied by Küçükgüçlü, Esen, and Yener (2009) (Cronbach's $\alpha = 0.92$).

Beck Depression Inventory. The Beck Depression Inventory (BDI) was used to evaluate symptoms of depression. Higher scores indicate more serious symptoms of depression. The inventory was previously translated into Turkish and validated for a Turkish population (Öner, 1997). The BDI is a 21-item scale, with a maximum score of 63.

World Health Organization Quality of Life Measure–Brief Version. The World Health Organization Quality of Life Measure–Brief Version (WHOQOL-BREF) was translated into Turkish by Eser and Baydur (2007). High scores indicate a better QOL (total score range = 27 to 135). The Turkish version (Eser & Baydur, 2007) has high content validity and satisfactory internal consistency.

Neuropsychiatric Inventory Questionnaire. The Neuropsychiatric Inventory Questionnaire (NPI) is an instrument used to assess behaviors commonly observed in individuals with dementia (Cummings et al., 2002). The NPI is a 12-item scale with total scores ranging from 12 to 144. Cronbach's α was 0.79 for the Turkish version of the NPI (Akça, Kalem, Hanagası, Cummings, & Gürvit, 2005).

The Quality of Life Scale for Patients with Alzheimer's Disease. The QOL-AD is an instrument used to assess QOL of individuals with dementia. The scale contains 13 items, with scores ranging from 13 to 52. Cronbach's α was 0.84 for the Turkish version of the scale and the intraclass correlation coefficient was 0.79. However, Cronbach's α was 0.77 for caregivers' reports on patients' QOL and the intraclass correlation coefficient was 0.72 (Akpınar & Küçükgüçlü, 2012).

Data Analysis

For analysis of all pre- and posttest data, SPSS version 15.0 for Windows was used. Statistical tests were generally considered significant at $p < 0.05$, but as the comparison involved three pairs (pretest–posttest 1, pretest–posttest 2, posttest 1–posttest 2), the new significance level was divided into three (Akgül, 2005) and determined to be 0.0166. Homogeneity of the two study groups was examined by comparing the sociodemographic variables and baseline scores with dependent variables (e.g., CBI, BDI, WHOQOL-BREF, QOL-AD) between groups using chi-square or Student's t tests, as appropriate. Repeated measures analysis of variance (ANOVA) was used to test for significant differences between caregivers and patients in the experimental and comparison groups, and for the major variables of caregiver burden, caregiver depression, caregiver QOL, NPI, and QOL-AD (group $\times$ time). In addition, ANOVA was used to test differences between caregivers and patients in the experimental and comparison groups at pre-intervention, post-intervention, and 6-month follow up.

Because data were missing for five participants (two in the experimental group and three in the comparison group), an intention-to-treat analysis was conducted

on data from participants who had a baseline measurement and had received at least one treatment (Whitaker, Sutton, & Burton, 2006), thus maintaining the advantages of random assignment and enhancing validity of the study findings. Bonferroni correction *t* tests for independent groups were used for further analysis of the difference between the two groups, and one-way ANOVA was used for comparison.

RESULTS

Registered patient lists from an outpatient clinic at a University hospital and the Alzheimer's Association were screened for eligibility and 70 individuals with dementia and their caregivers were randomized for treatment (Figure). In the experimental and comparison groups, caregivers were mainly spouses ($n = 18$ [52%] versus $n = 17$ [49%]). In both groups, two thirds of caregivers were female and most had completed high school. The mean age of caregivers in the experimental group was 61.2 years ($SD = 13.14$ years), and 62.28 years ($SD = 12.76$ years) in the comparison group. Table 2 presents the main sociodemographic characteristics of the sample. Using *t* tests or chi-square tests, no statistically significant differences were found between the two groups regarding sociodemographic characteristics or mean scores on the dependent variables at baseline (Table 2 and Table 3).

It was hypothesized that the PLST-based intervention would be more effective than routine care in reducing burden and depression, and in increasing QOL of caregivers of individuals with dementia. The first hypothesis was not supported by the research findings. There were no statistically significant differences between the two study groups at any of the data collection points (Table 3). There was improvement at 3- and 6-month follow up in only the

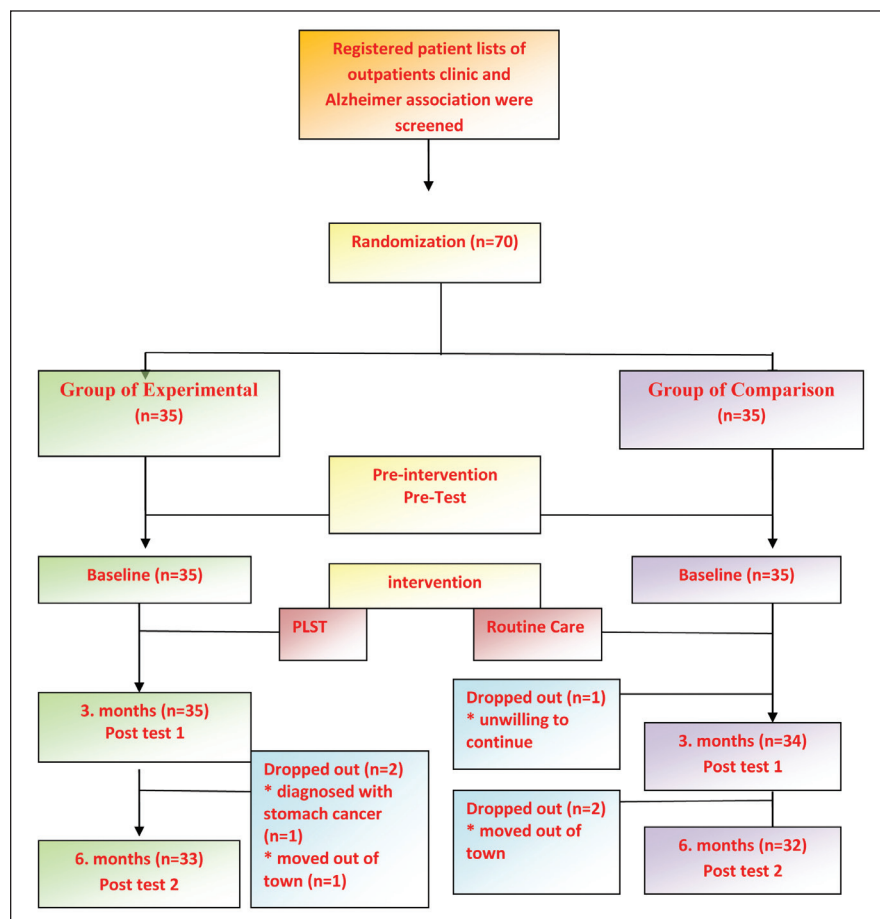


Figure. Study design comparing the Progressively Lowered Stress Threshold (PLST) model and routine care.

experimental group (Table 4). CBI scores in the experimental group decreased from baseline (38.02 to 33.33) and then remained stable, indicating less burden over time, whereas burden scores were lower initially but increased over time for the comparison group, indicating a growing sense of burden. BDI scores in the experimental group decreased from baseline (9.74 to 7.46) and then remained stable, whereas they increased over time for the comparison group, indicating increasing symptoms of depression. WHOQOL-BREF scores in the experimental group increased from baseline (86 to 89.91), indicating better QOL, but decreased over time for the comparison group.

It was also hypothesized that the PLST-based intervention would be

more effective than routine care in reducing distressing behavioral symptoms and in increasing QOL for individuals with dementia. The second hypothesis was also not supported by the research findings. There were no statistically significant differences between the two study groups on any of the variables for individuals with dementia (Table 4). There was improvement at 3- and 6-month follow up in only the experimental group (Table 4). QOL-AD scores in the experimental group increased from baseline (31.6 to 33.6), indicating better QOL, but were stable over time for the comparison group. The mean scores and results of the ANOVAs for the dependent variables in the repeated measures, when considered separately, are summarized in Table 4.

TABLE 2

SOCIODEMOGRAPHIC CHARACTERISTICS OF THE SAMPLE (N = 70)

Variable	PLST (n = 35)	Routine Care (n = 35)	Chi-Square/t Test (p Value)
Age of care recipients (years) (mean, SD)	78.6 (5.67)	76.29 (5.76)	$t = 1.675$ (0.09)
Age of caregivers (years) (mean, SD)	61.2 (13.14)	62.28 (12.76)	$t = -0.351$ (0.798)
Care recipient gender			
Female	22 (63)	19 (54)	$\chi^2 = 0^a$ (1)
Male	13 (37)	16 (46)	
Caregiver gender			
Female	26 (74)	25 (71)	$\chi^2 = 0.53$ (0.467)
Male	9 (26)	10 (29)	
Care recipient education level			
Primary school	24 (69)	17 (49)	$\chi^2 = 4.773$ (0.189)
High school	5 (14)	12 (34)	
University	6 (17)	6 (17)	
Caregiver education level			
Primary school	13 (37)	11 (31)	$\chi^2 = 0.452$ (0.798)
High school	16 (46)	16 (46)	
University	6 (17)	8 (23)	
Relationship of caregiver to care recipient			
Spouse	18 (52)	17 (49)	$\chi^2 = 0.789$ (0.674)
Child	11 (31)	14 (40)	
Daughter-in-law	6 (17)	4 (11)	
Monthly household income			
Equal to expenditure	33 (94)	33 (94)	(0.693) ^b
More than expenditure	2 (6)	2 (6)	
Duration of caregiving			
<1 year	2 (6)	6 (17)	$\chi^2 = 2.587$ (0.274)
1 to 5 years	27 (77)	22 (63)	
≥6 years	6 (17)	7 (20)	

Note. PLST = Progressively Lowered Stress Threshold.

^a Yates correction was calculated because $n < 25$.

^b Fisher's exact test was performed because the expected number was < 5 .

DISCUSSION

The process of caring for patients with dementia may affect caregivers for many years, change the course of their lives, and cause them to experience burden, depression, and decreased QOL.

Outcome variables for the current study are sometimes thought to be impossible to alleviate because they are related to caregivers' perceptions of their physical, social, and psychological well-being, as well as their sense of satisfaction in daily

life. Because these variables are affected by a number of conditions, a gradual improvement or even a leveling out of burden, depression, and QOL (which often worsen in the process of care) could indicate clinical efficacy of the intervention.

TABLE 3**COMPARISON OF DEPENDENT VARIABLES BETWEEN GROUPS AT BASELINE**

Variable	Mean (SD)		t Test (p Value)
	PLST (n = 35)	Routine Care (n = 35)	
Caregivers' burden	38.02 (21.49)	31.19 (16.54)	1.49 (0.141)
Caregivers' depression	9.74 (4.43)	8.62 (3.18)	1.208 (0.231)
Caregiver WHOQOL-BREF	86 (9.15)	87.2 (5.69)	-0.659 (0.512)
Caregiver reported QOL-AD	31.6 (4.45)	32.83 (4.35)	-1.167 (0.247)
NPI of care recipient	17.03 (12.17)	14.51 (10.43)	0.928 (0.357)
Patient reported QOL-AD	37.14 (4.31)	37.46 (2.86)	0.36 (0.72)

Note. PLST = Progressively Lowered Stress Threshold; WHOQOL-BREF = World Health Organization Quality of Life Measure–Brief Version; QOL-AD = Quality of life–Alzheimer's disease; NPI = Neuropsychiatric Inventory Questionnaire.

TABLE 4**DEPENDENT VARIABLE SCORES AT PRE- AND POSTTEST AND REPEATED MEASURES ANOVA RESULTS^a**

Instrument	Mean (SD)						ANOVA
	Experimental Group			Comparison Group			
	Pretest	Posttest 1	Posttest 2	Pretest	Posttest 1	Posttest 2	
CBI	38.02 (21.49)	33.33 (18.37)	34.71 (18.82)	31.19 (16.54)	32.86 (16.1)	35.66 (17.07)	23.71*
BDI	9.74 (4.43)	7.46 (3.55)	7.97 (3.42)	8.63 (3.18)	8.83 (3.34)	9.86 (3.47)	31.43*
WHOQOL-BREF	86 (9.15)	89.91 (8.75)	88.51 (8.28)	87.2 (5.69)	86.2 (5.21)	85.23 (5.68)	23.22*
NPI	31.6 (4.45)	33.6 (4.72)	33.48 (4.69)	32.83 (4.35)	32.63 (4.64)	31.91 (4.3)	28.78*
QOL-AD	31.6 (4.45)	33.6 (4.72)	33.48 (4.69)	32.83 (4.35)	32.63 (4.64)	31.91 (4.3)	20.19*

Note. ANOVA = analysis of variance; CBI = Caregiver Burden Inventory; BDI = Beck Depression Inventory; WHOQOL-BREF = World Health Organization Quality of Life Measure–Brief Version; NPI = Neuropsychiatric Inventory; QOL-AD = Quality of life–Alzheimer's disease.

^a Pretest = baseline measurement before assignment into study groups; posttest 1 = 3 months after intervention; and posttest 2 = 6 months after intervention.

*p < 0.001.

Although some interventional studies and meta-analyses have failed to find statistically significant results, the clinical benefits of interventions have been emphasized (Brodaty & Arasaratnam, 2012; Ostwald, Hepburn, Caron, Burns, & Mantell, 1999; Schulz et al., 2002). These studies reported a number of variables that may have affected outcomes, including: the interventions comprised short

training programs; the research used different follow-up processes; sample sizes were insufficient; and variables examined were lower in baseline values. All of these factors may have led to the failure to detect changes in the expected direction.

Similarly, as the current study applied no cutoffs regarding NPI score as part of inclusion criteria, both groups comprised patients who had good NPI scores

as a result of randomization. It is possible that the expected decline after the intervention did not occur because the patients already had a low (i.e., good) NPI score at baseline. Another reason for lack of statistical differences between groups at follow up may be related to difficulties of families regarding the application of the PLST-based intervention in the home. Due to cultural expectations of privacy

in the domestic life in Turkey, the researchers (B.A.S., Ö.K.) were unable to observe whether the care plan was being effectively applied by caregivers of the patient in the home. Thus, data relied on caregivers' statements.

Patients received care from partners, children, daughters-in-law, and other relatives. Anecdotally, it was observed that caregivers who were the daughters of patients had greater success in learning and applying the PLST-based intervention, whereas sons and partners experienced difficulties in adhering to intervention protocol. Although contact was established with caregivers in person

The systematic interventions of Olazarán et al. (2010) aimed at behavioral problems and comprised interventions lasting approximately 30 minutes 3 days per week for at least 20 weeks. The duration and frequency of interventions applied were thus longer and more frequent compared to the intervention in the current study. In other studies, the duration of the PLST-based intervention has varied from 3 (Huang, Shyu, Chen, Chen, & Lin, 2003), 12 (Gerdner, Buckwalter, & Reed, 2002; Stolley, Reed, & Buckwalter, 2002), and 16 (Garand et al., 2002) months. Results of the current study show that the model is effective

IMPLICATIONS

Although some studies in other countries have supported the efficacy of the PLST-based intervention for managing behavioral problems, there have been few randomized controlled studies and none using the model in Turkey. Turkish culture and society traditionally promote strong family bonds and familial care for those in need; individuals and society in general find it culturally difficult to accept institutional care. Support for individuals who provide home care is thus required. The PLST-based intervention is valuable in providing support for home care. Adapting the PLST to current routine care in Turkey is not possible due to the limited time of physicians and clinics. However, it is possible to assign nurses to train caregivers in the clinic. Nurses could then provide education and training for the PLST-based intervention to better meet caregivers' needs. Thus, the current routine care could be revised in this context, potentially improving caregiver burden, mood, and QOL.

RECOMMENDATIONS

The current intervention was applied for only 6 months due to time and cost constraints. Future studies with longer intervention and follow up may help evaluate long-term effects of PLST-based intervention. Adding a qualitative component (i.e., an individualized interview) may help provide a better understanding of caregivers' experiences with and effects of the PLST-based intervention.

CONCLUSION

The PLST-based intervention has been used in many experimental research projects in different countries (Huang et al., 2003; Smith, Gerdner, Hall, & Buckwalter, 2004; Stolley et al., 2002), but not in Turkey. The current study was thus the first randomized con-

Turkish culture and society traditionally promote strong family bonds and familial care for those in need; individuals and society in general find it culturally difficult to accept institutional care.

and by telephone throughout the study, and problems with the care plan and solutions were discussed, lack of expected statistical differences on NPI scores may have been caused by problems that were neither expressed by caregivers nor observed by the researcher during application of the intervention—this may be why the expected differences between the groups regarding caregiver burden, depression, and QOL were not attained.

Although there were no significant differences between the groups, caregivers in the experimental group showed more improvement in each of the variables (i.e., burden, depression, and QOL) than those in the comparison group. These results may be related to positive changes in perceptions of caregivers to the caregiving process, mood, and QOL. Support of caregivers using principles of the PLST-based intervention at home may promote changes in their perceptions.

tive in short- and long-term applications. A 6-month intervention was planned for the current study for reasons of time and cost effectiveness.

LIMITATIONS

Observations indicated that the PLST-based intervention was generally tolerated well by caregivers and individuals with dementia. Daughters of patients were generally more careful than sons in adhering to intervention protocol. It was difficult to set fixed time limits for home visits, as caregivers in both groups generally wanted to offer extended hospitality and refreshments. Many caregivers were spouses and it proved more difficult to teach the PLST-based intervention to these older adult caregivers. Some caregivers (especially males) also stated that altering care arrangements would not cause a difference in the patient's condition.

trolled trial in Turkey of the PLST-based intervention for caregivers and individuals with dementia. The PLST-based intervention was clinically effective in decreasing burden and depression levels of family members caring for individuals with Alzheimer's disease and in increasing their QOL. However, there was no change in behavioral problems in individuals with dementia and no statistically significant differences were found between the two groups.

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